Roll No. _____

Thapar Institute of Engineering and Technology
Electronics and Communication Department
End Semester Test

Course Name: Database Management Systems
Date & Day: 10th December 2025, Wednesday
Name of Faculty: Dr. Ashu Sharma, Dr. Deepak Rakesh, Dr. Poonam Verma Bhardwaj

Course Code: UEC716/ UCS310
Time: 2:00 - 5:00 PM
Max Marks: 50

Note: Attempt any five questions, Assume any missing value.

PLEASE ATTEMPT ALL SUBPARTS OF A QUESTION AT ONE PLACE.

Q.No.	Questions	Marks	CL	BL
Q1	Describe the following terms with proper examples: (a) Schema (b) Weak Entity (c) Is a relationship (d) Data Abstraction (e) Overlapping and disjoing constraint	10	C1, C2	L2
Q2	Consider the schema of a banking database given below. branch (branch_name, branch_city, assets) customer (customer_name, customer_street, customer_city) loan (loan_number, branch_name, amount) borrower (customer_name, loan_number) account (account_number, branch_name, balance) depositor (customer_name, account_number) Express the following queries using relational algebra: i. Find loan number of those loans with loan amount between INR 90,000 and 100,000. ii. Find names of all branches that have greater assets than some branch located in Delhi. iii. Find all customers who have an account but no loan. iv. Find the number of depositors in the bank. v. Find all loan number for loans made at the Patiala branch with loan amounts greater than INR 10000.	10	C3	L3
Q3	Express the queries given in Q.3 in equivalent SQL format	10	C5	L3
Q4	Consider the following schedule for transactions T1, T2, and T3: T1: Read (X); Write (X) T2: Read (Y); Write (Y); Read (X); Write (X) T3: Read (Y); Write (X) (Use the interleaved schedule given above for transactions T1, T2 and T3.) (a) Construct the complete conflict (precedence) graph by identifying all read–write, write–read, and write–write conflicts among T1, T2, and T3 on data items X and Y.	4+2+4	C4	L4, L5

	(b) Based on the precedence graph, determine whether the schedule is conflict serializable. If it is serializable, identify all possible equivalent serial schedules. (c) Out of the possible serial orders you identified, determine which single serial schedule exactly reflects the execution ordering imposed by the given interleaving. Justify your answer.			
Q5	Database pages initially: P1=300, P2=600. Log records (in order): L1: <T1 start> L2: <T1, write, P1, 300, 320> L3: <T2 start> L4: <T2, write, P2, 600, 650> L5: <T1, write, P2, 650, 700> L6: <T3 start> L7: <T3, write, P1, 320, 350> L8: <T2 commit> L9: <system crash> (a) Identify which transactions are committed and which are uncommitted at the instant of the crash. (b) Perform the REDO phase. Apply the logged updates (in appropriate order) and compute the final values of P1 and P2 after REDO. Show all steps. (c) Perform the UNDO phase and give the final P1, P2 values after UNDO. Show all steps. Note: Perform parts (b) and (c) separately:	2+4+4	C4	L3, L4
Q6	Consider the relation R(A, B, C, D, E, F, G, H) with the following set of functional dependencies: • CH → G • A → B C • B → C H F • E → A • F → E G Assume that F^+ is exactly the complete set of functional dependencies that hold for R. (a) Compute the candidate key(s) of R. Show the attribute-closure steps clearly. (b) Using the candidate key(s) found in (a), evaluate each given FD and state whether it violates the conditions for: • 2NF, • 3NF, and • BCNF. (c) Based on your analysis, state the highest normal form satisfied by R and justify your conclusion concisely	3+5+2	C3	L3, L5

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